TRƯỜNG ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Probability Theory and Random Variables

Mathematical Foundations for Stochastic Processes and Filtering Theory

Chapter 2 · Lab discussion

ONE LOVE. ONE FUTURE.

From uncertainty to state estimation

How can noisy observations reveal an unobserved state?

Probability space › Random variables › Joint distributions › Conditional probability › Gaussian conditioning

Represent uncertainty **Describe dependence** **Update knowledge**

X (hidden state) $\longrightarrow$ Y (noisy observation)

A filter learns about X by conditioning its law on observations.

A random variable is not its realized value

Random variable	Realization	Distribution
$X : \Omega \to \mathbb{R}$	$x = X(\omega)$	$p_X(x)$
A measurable map before observation	One numerical value after realization	A law assigning probability to sets

$$m_X = \mathbb{E}[X], \qquad \sigma_X^2 = \mathbb{E}[(X - m_X)^2]$$

Mean describes location; variance describes squared dispersion.

A distribution describes uncertainty across possible realizations.

Moments summarize; characteristic functions encode

Selected distribution features

$$\mu_n' = \mathbb{E}[X^n]$$

Mean: location
Variance: dispersion
Higher moments: shape features

A complete Fourier description

$$\varphi_X(u) = \mathbb{E}[e^{iuX}]$$

Always exists and determines the law.
Moments require integrability.
Independent sums give products.

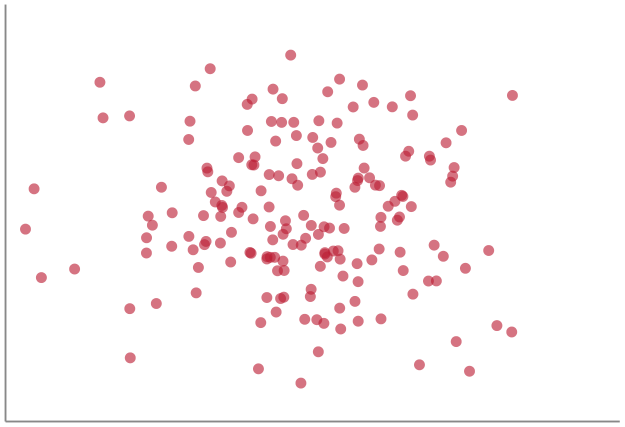
$$X \sim \mathcal{N}(m, \sigma^2) \quad \Rightarrow \quad \varphi_X(u) = e^{ium - \frac{1}{2}\sigma^2 u^2}$$

Mean and variance alone generally do not determine a distribution.

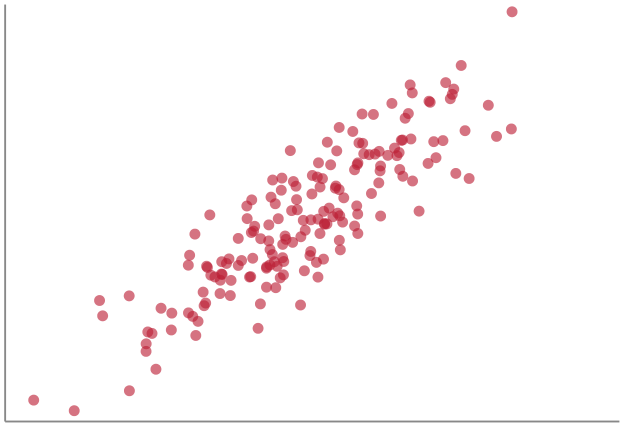
Joint laws carry dependence that marginals omit

Two pairs can have the same standard Gaussian marginals.

Independent components



Dependent components



$$p_X(x) = \int p_{X,Y}(x,y)\, dy$$

Marginalization removes a variable; it does not reveal dependence.

Covariance is directional uncertainty

$$P_X = \mathbb{E}[(X - m_X)(X - m_X)^T]$$

$$P_X = \begin{bmatrix} \sigma_1^2 & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \sigma_2^2 \end{bmatrix}$$

Diagonal: component variances
Off-diagonal: cross-covariances

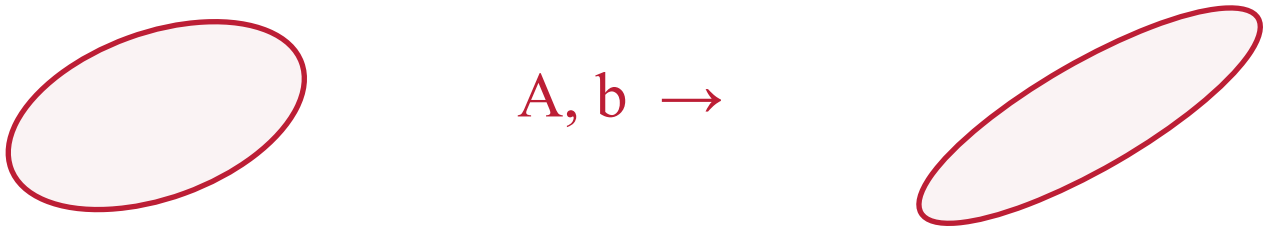
Symmetric
Positive semidefinite

$$a^T P_X a = \mathrm{Var}(a^T X) \geq 0$$

Every direction has a nonnegative variance; no Gaussian assumption.

Affine maps propagate mean and covariance

$$Y = AX + b$$



$$m_Y = Am_X + b$$

$$P_Y = AP_XA^T$$

These moment identities need finite second moments, not Gaussianity.

If X is Gaussian, Y is also Gaussian—even for a singular map A.

Linear state prediction is an affine transformation of uncertainty.

Bayes' theorem turns observations into information

$$p_{X|Y}(x \mid y) = \frac{p_{Y|X}(y \mid x)\, p_X(x)}{p_Y(y)}$$

Prior	Likelihood	Posterior
Uncertainty before the observation	Compatibility of a state with the observed value	Uncertainty after the observation

The evidence normalizes the posterior to integrate to one.

Bayesian updating applies to general distributions, not only Gaussians.

Conditional expectation is an estimator

$$\widehat{X} = g(Y) = \mathbb{E}[X \mid Y], \qquad \hat{x} = g(y)$$

Before observation: random estimator. After Y = y: numerical estimate.

Remaining uncertainty

$$P_{X|Y} = \mathbb{E}[(X - g(Y))(X - g(Y))^T \mid Y]$$

$$P_X = \mathbb{E}[P_{X|Y}] + P_{g(Y)}$$

With finite second moments, g(Y) minimizes expected squared error among all estimators based on Y. This does not require Gaussianity.

Conditional covariance measures uncertainty remaining after observation.

Joint Gaussian structure closes the model

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} m_X \\ m_Y \end{bmatrix}, \begin{bmatrix} P_X & P_{XY} \\ P_{YX} & P_Y \end{bmatrix} \right)$$

Theorem 2.10 Zero cross-covariance implies independence.

Theorem 2.11 Affine transformations preserve Gaussianity.

Theorem 2.12 Every marginal subvector remains Gaussian.

Gaussian marginals alone do not establish a jointly Gaussian model.

Gaussian conditioning is an innovation correction

Assume joint Gaussianity and an invertible observation covariance.

$$K = P_{XY}P_Y^{-1}, \qquad \nu = y - m_Y$$

Gain × innovation shifts the prior mean

$$m_{X|y} = m_X + K(y - m_Y)$$

$$P_{X|Y} = P_X - P_{XY}P_Y^{-1}P_{YX}$$

The covariance is independent of the observed value. $0 \preceq P_{X|Y} \preceq P_X$

$$R = X - m_X - K(Y - m_Y), \qquad R \perp Y$$

The independent Gaussian residual explains the remaining uncertainty.
For singular observation covariance, retain nonredundant coordinates.

A noisy measurement moves and sharpens the prior

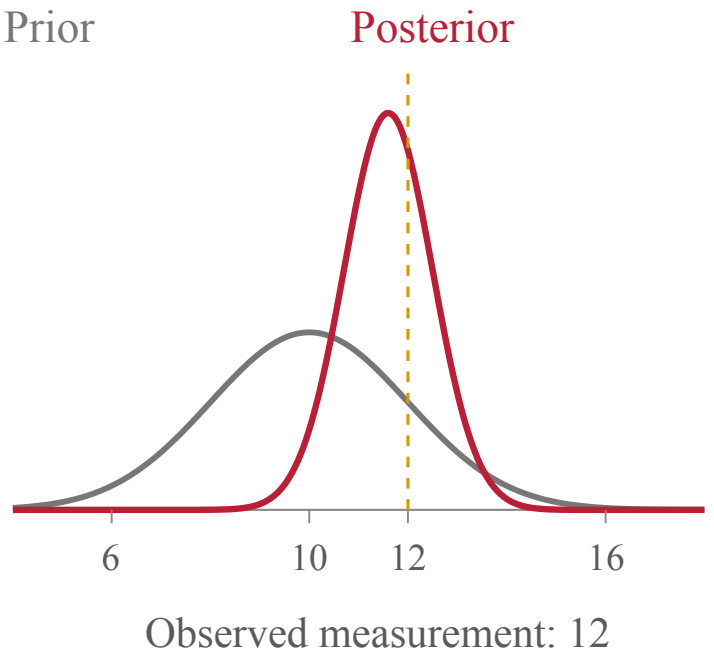
$$X \sim \mathcal{N}(10,4), \quad Y = X + V, \quad V \sim \mathcal{N}(0,1), \quad y = 12$$

$$m_Y = 10, \quad P_Y = 5, \quad P_{XY} = 4$$

$$K = \frac{4}{5} = 0.8$$

$$m_{X|12} = 10 + 0.8(2) = 11.6$$

$$P_{X|Y} = 4 - \frac{16}{5} = 0.8$$



Posterior mean 11.6, variance 0.8: prior and measurement are combined.

Gaussian closure gives the Kalman recursion

$$X_k = FX_{k-1} + W_{k-1}, \qquad Y_k = HX_k + V_k$$

Prediction · Theorem 2.11

$$m_k^- = Fm_{k-1}^+$$
$$P_k^- = FP_{k-1}^+F^T + Q$$

− predicted; + updated

Update · Theorem 2.13

$$K_k = P_k^- H^T S_k^{-1}$$
$$m_k^+ = m_k^- + K_k(y_k - Hm_k^-)$$
$$P_k^+ = P_k^- - K_k S_k K_k^T$$

$$S_k = HP_k^- H^T + R$$

Gaussian initial state; initial state and all noises mutually independent.
Zero-mean Gaussian noises; Q, R their covariances; innovation covariance invertible.

Bayesian filtering is broader than Gaussian filtering

$$p(x_k \mid y_{1:k-1}) = \int p(x_k \mid x_{k-1}) p(x_{k-1} \mid y_{1:k-1})\, dx_{k-1}$$

$$p(x_k \mid y_{1:k}) \propto p(y_k \mid x_k) p(x_k \mid y_{1:k-1})$$

General Bayesian filtering

Propagate a conditional law.
Gaussianity is unnecessary.
General models may need numerical or approximate representations.

Linear Gaussian / Kalman

Propagate a mean and covariance.
The Gaussian posterior is exact.
Closure requires the additional linear Gaussian model assumptions.

Finite parameter closure is a special structure, not a Bayes requirement.

The foundation: represent, connect, condition

01 **Represent uncertainty**

A distribution is the mathematical model of what is unknown.

02 **Describe dependence**

Joint laws connect hidden states with observable quantities.

03 **Update knowledge**

Conditioning turns observations into posterior information.

Random variables → Joint laws → Conditioning → Gaussians → Filtering

Gaussian closure turns this foundation into an exact Kalman recursion.